



Why Accident Severity Prediction Matters

Dataset: “US Accidents (2016–2023)”

Why Accident Severity Prediction Matters

Every year, millions of traffic accidents occur across the United States, but their severity varies dramatically. Understanding and **predicting severity levels is crucial for effective emergency response** and resource allocation.

High-severity crashes cause:

- Longer traffic delays and extensive congestion
- Higher risk of serious injury or fatality
- Greater strain on emergency responders and medical facilities

Anticipating accident severity enables transportation agencies to proactively allocate resources, implement targeted safety interventions, and improve overall road safety outcomes for communities nationwide.



Emergency Response



Traffic Management



Delay Reduction



Research Questions & Stakeholders

Research Questions

This project addresses three core questions to advance traffic safety prediction:

1. Can we accurately predict the **severity level** (1–4) of an accident using information available at the time and location of the crash?
2. Which environmental and temporal conditions, such as weather patterns, time of day, or road distance affected, are most strongly associated with **high-severity crashes** (levels 3–4)?
3. Are there **identifiable "high-risk situations"** that combine multiple factors, such as nighttime conditions plus adverse weather plus extended road impact?

Key Stakeholders

Our findings benefit multiple groups invested in road safety and traffic management:



Transportation Departments

Better information to plan safer roads and reduce severe crashes



Urban Planners

Reliable data to guide long-term, safer city and roadway designs



Emergency Services

Improve preparedness and response time to accidents



Insurance Companies

Improve understanding of crash risk to manage costs and promote safety



General Public

Experience safer driving conditions, less traffic from accident delays, quicker response times to accidents, and less accidents/fatalities overall

US Accidents (2016–2023) – Kaggle Dataset

Our analysis utilizes the comprehensive **US Accidents (2016–2023)** dataset, publicly available on Kaggle and curated by Sobhan Moosavi. This dataset represents one of the most extensive collections of traffic accident data available for research purposes.

7.7M

Accident Records

Comprehensive coverage of traffic incidents

49

U.S. States

Contiguous states (excludes Alaska & Hawaii)

7

Years of Data

February 2016 to March 2023

The dataset was collected in **real time** through **multiple traffic APIs**, capturing incidents as they occurred across the nation. Data were aggregated from various sources to provide comprehensive spatial and temporal coverage resulting in **46 features**.

Data Collection: Real-Time Traffic APIs

The US Accidents dataset leverages a sophisticated multi-source data collection infrastructure that captures traffic incidents in real time across the United States. This approach provides comprehensive coverage beyond traditional police crash reports.

Data Source Architecture



Critically, this dataset focuses on **incidents that affect traffic flow**, not merely police-reported crashes. This broader scope captures the real-world impact of accidents on transportation networks, making it ideal for severity prediction models.

Severity and Key Features from Kaggle

Target Variable: Severity

The severity rating represents the **traffic impact** of each accident, measured on a numerical scale from 1 to 4:

01

Minimal Impact

Short delays, minimal traffic disruption

02

Minor Impact

Most common class; moderate delays

03

Serious Disruption

Significant traffic impact and delays

04

Major Impact

Severe delays, extensive road closure

Core Dataset Features

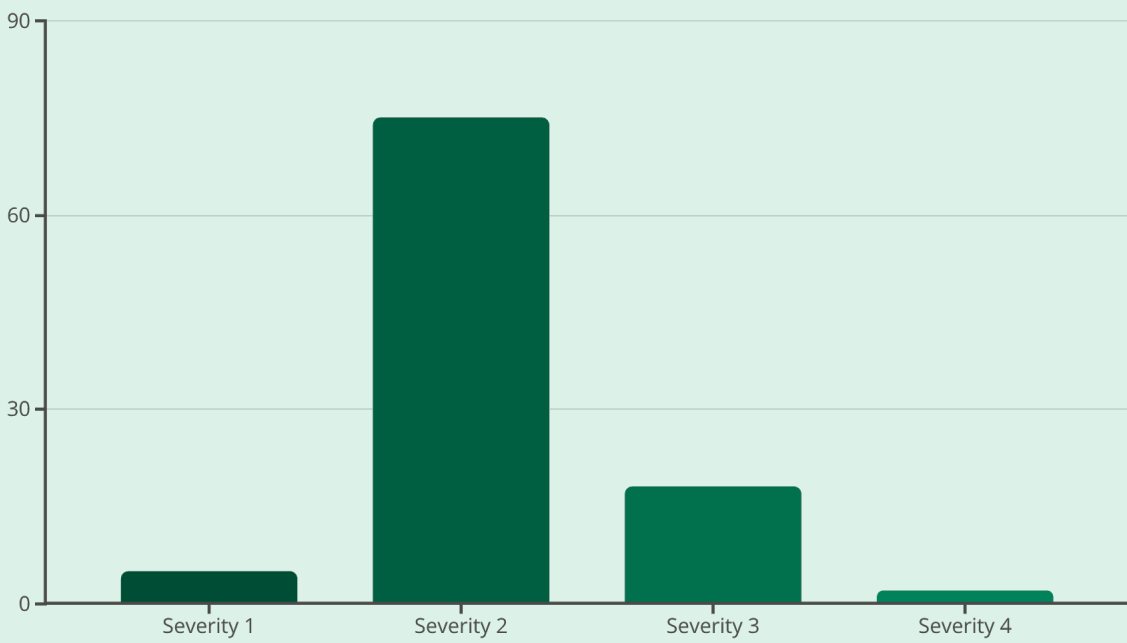
The Kaggle file (US_Accidents_March23.csv) contains rich information for each accident record:

Column Name	Description
ID	Unique accident identifier
Source	Data provider/API source
Start_Time, End_Time	Incident timestamps (local time)
Start_Lat, Start_Lng	GPS coordinates (incident start)
End_Lat, End_Lng	GPS coordinates (incident end)
Distance(mi)	Length of road affected

Severity Distribution & Time Span

Severity Distribution

The dataset exhibits a **strong class imbalance** that presents both challenges and opportunities for modeling:

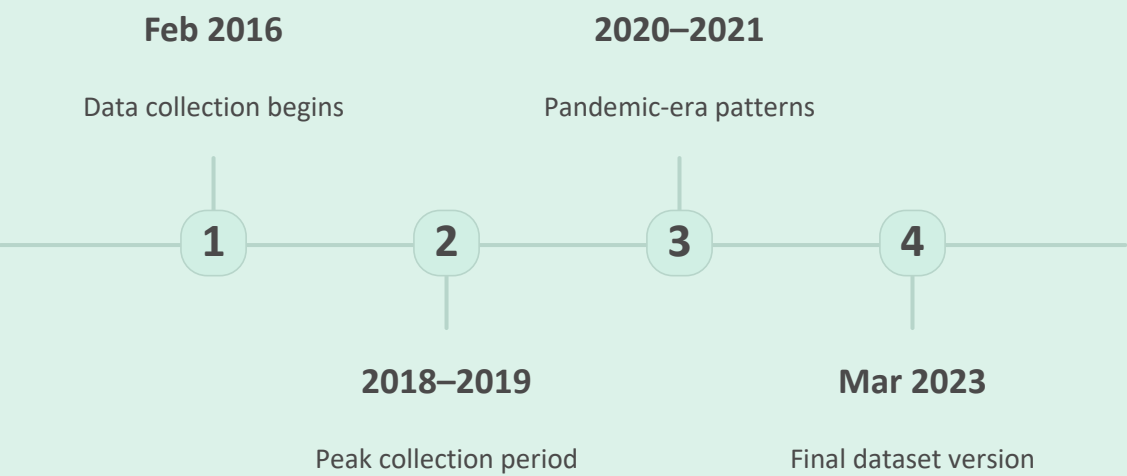


Severity 2 dominates the dataset, representing the vast majority of recorded accidents. **Severity 3** forms a smaller but meaningful minority, while **Severity 4**, the most critical for traffic management is the rarest.

This imbalance means naive models tend to predict Severity 2 by default, requiring specialized techniques to identify high-severity events.

Temporal Coverage

The dataset spans approximately **7 years and 2 months** of continuous accident data collection:



While coverage is extensive, some days may be missing due to API connectivity issues or network outages. The multi-year span enables analysis of seasonal patterns, long-term trends, and the impact of changing traffic conditions over time.

Features Used and Feature Engineering

Our modeling approach leverages both numerical and categorical features to capture the complex relationships between environmental conditions, temporal factors, and accident severity.

Numeric Features

- **Start_Lat, Start_Lng:** Geographic coordinates for approximate location
- **Distance(mi):** Length of road segment impacted by the incident
- **Temperature(F):** Ambient temperature at accident time
- **Humidity(%):** Relative humidity levels
- **Pressure(in):** Atmospheric pressure measurement
- **Visibility(mi):** Visibility distance at incident location
- **Wind_Speed(mph):** Wind speed during the accident

Categorical Features

- **Weather_Condition:** Descriptive weather status (Clear, Rain, Snow, Fog, etc.)
- **Sunrise_Sunset:** Binary indicator for Day versus Night conditions

Preprocessing Pipeline



Modeling Approach & Imbalance Handling

We frame accident severity prediction as a **supervised multiclass classification problem**, where the goal is to predict one of four severity levels based on available features at the time of the incident.

Addressing Class Imbalance with SMOTE

The dominant presence of Severity 2 cases creates a significant modeling challenge. Without intervention, models would simply predict Severity 2 for nearly all cases, achieving high accuracy but poor practical utility.

We apply **SMOTE (Synthetic Minority Over-sampling Technique)** to the training set only. SMOTE generates synthetic examples of minority classes by interpolating between existing samples, creating a more balanced training distribution while preserving the original test set for realistic evaluation.

Models Evaluated

1

Logistic Regression

Baseline linear model providing interpretable coefficients and fast training. Serves as performance benchmark for more complex approaches.

2

Random Forest

Ensemble of decision trees capturing nonlinear relationships and feature interactions without extensive hyperparameter tuning.

3

XGBoost

Gradient boosting framework optimized for tabular data. Expected to deliver strongest performance through iterative error correction.

Model Performance & What Matters Most

Performance Overview

Our models demonstrate varying capabilities in predicting accident severity, with clear differences in performance across the three approaches tested:

1) Random Forest:

- Best-performing model
- Highest accuracy (0.77)
- Best weighted F1-score (0.77)
- Most consistent predictions across major classes (Severity 2 & 3)

2) XGBoost:

- Performed very competitively to Random Forest
- Accuracy (0.76) and weighted F1 very close to Random Forest
- Strong on major classes but unstable on rare classes (Severity 1 & 4)

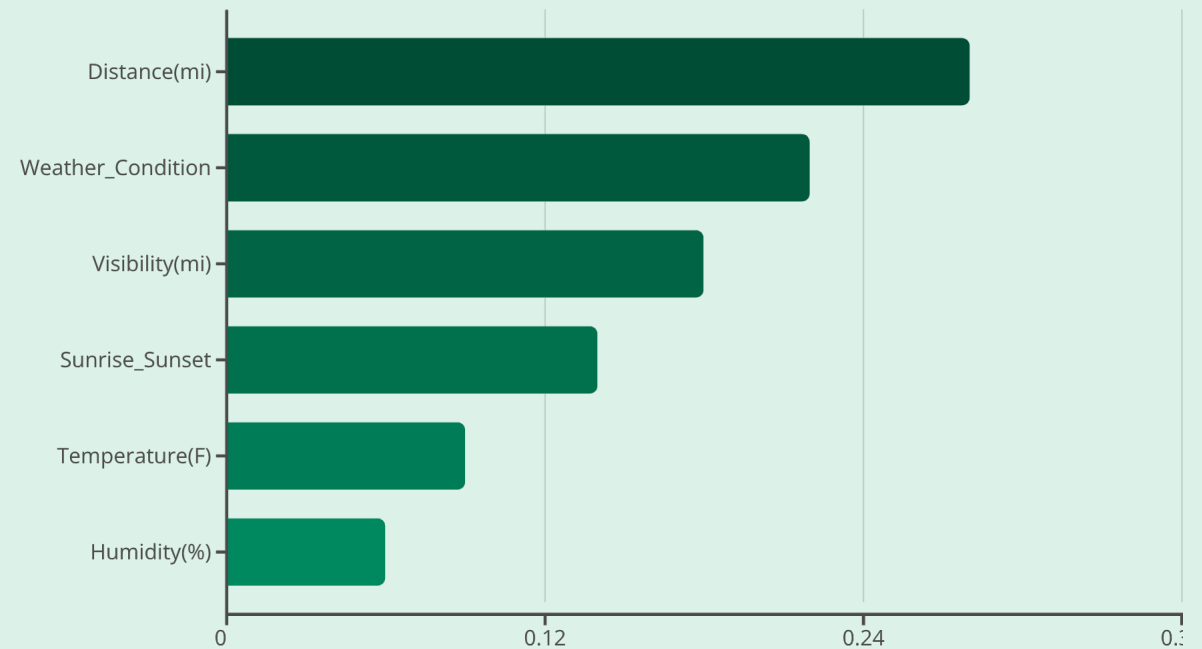
3) Logistic Regression:

- Performed very poorly
- Accuracy only 0.22
- Struggled due to nonlinear relationships and extreme class imbalance

Rare severity levels (1 & 4) remained difficult and were poorly predicted regardless of model since there were very few real examples (34 and 27 in test set)

Feature Importance (XGBoost)

Analysis of XGBoost's decision-making reveals which factors most influence severity predictions:



More severe traffic impacts tend to occur on **longer road distances**, during **adverse weather conditions** (heavy rain, snow, storms), during **reduced visibility**, and **at night**. Temperature and humidity also play supporting roles in severity determination.

Implications and Real-World Readiness

Implications

Key Findings: Environmental factors (road distance, weather, visibility, time of day) are strong predictors of crash severity. Severity can be estimated using information available *at the time and location of a crash*.

Transportation agencies could use similar models to:

- Flag high-risk situations in advance
- Identify signs, warnings, or speed adjustments
- Prioritize lighting, drainage, and road maintenance improvements

Real-World Readiness

While the model is promising, it is not fully ready for real-world deployment.

Key limitations:

- Underperforms on the most severe crashes
- Trained on only part of the dataset due to limited computing resources
- Missing key variables like speed limits and traffic volume
- Dataset is not updated in real time

These limitations are why the predictions are not yet stable, explainable, or continuously monitored.

Next Steps

To move toward **production readiness**:

- Train on the full dataset to improve performance on severe crashes
- Implement data quality checks for outliers and missing values
- Build a real-time pipeline that pulls in live traffic and weather data and produces crash-risk scores
- Add monitoring + alerts to detect when model behavior changes
- Evaluate fairness across states and urban/rural regions; re-train if disparities appear

Questions?